

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

BIOMOLECULES

LECTURE-6

ZOOLOGY

DAVOOD SIR (CEO of NG)



Physics Wallah

Topics To Be Covered

- 1 **Enzymes** 
 - lec-1 ✓
 - lec-2
- 2 **Previous Year NEET MCQ's**

Enzymes

Enzymes are the biocatalysts that increases the rate of a chemical reaction within the body.



Properties of an Enzyme

- ① Enzymes are generally 3rd proteins but some enzymes are made of RNA and are called Ribozymes
- ② Enzymes have crevices or pockets known as Active sites where substrate binds.
- ③ Enzymes are heat sensitive and function under a specific temp. If temp rises more than 40°C they get denatured and at low temp they become inactive.
- ④ Enzymes in thermophilic organisms (like in hot vents) are thermally stable and function at 80°C.
Heat + Love
- ⑤ Enzymes (Biocatalysts) greatly increase the rxn rates. E.g:



- Rxn Rate is 10-million times faster due to C.a enzyme.
- Carbonic anhydrase → fastest enzyme.

Properties of an Enzyme

Enzymes are highly powerful and specific, each catalysing a unique reaction.

Metabolic pathway: A series of enzyme-catalysed steps (same or different enzymes)e.g:-

Glucose + Oxygen → 2 Pyruvic acid + Water (via 10 enzyme-catalysed steps).

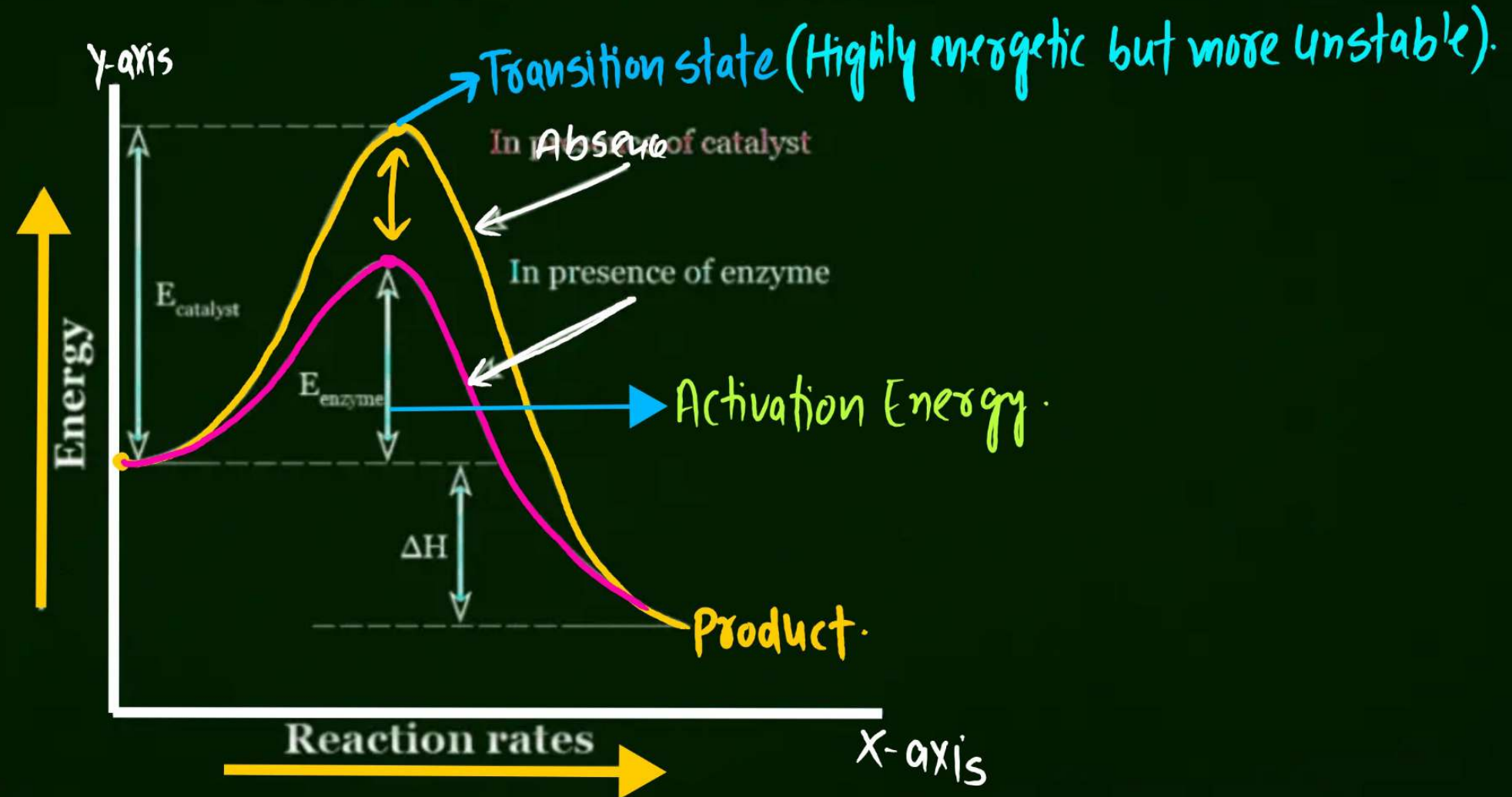
In aerobic conditions: pyruvic acid is formed.

In anaerobic muscles: lactic acid is formed.

In yeast (fermentation): Ethanol (alcohol) is formed.

Thus, same pathway can lead to different end products depending on conditions.

How Enzyme Fastens a Chemical Reaction?

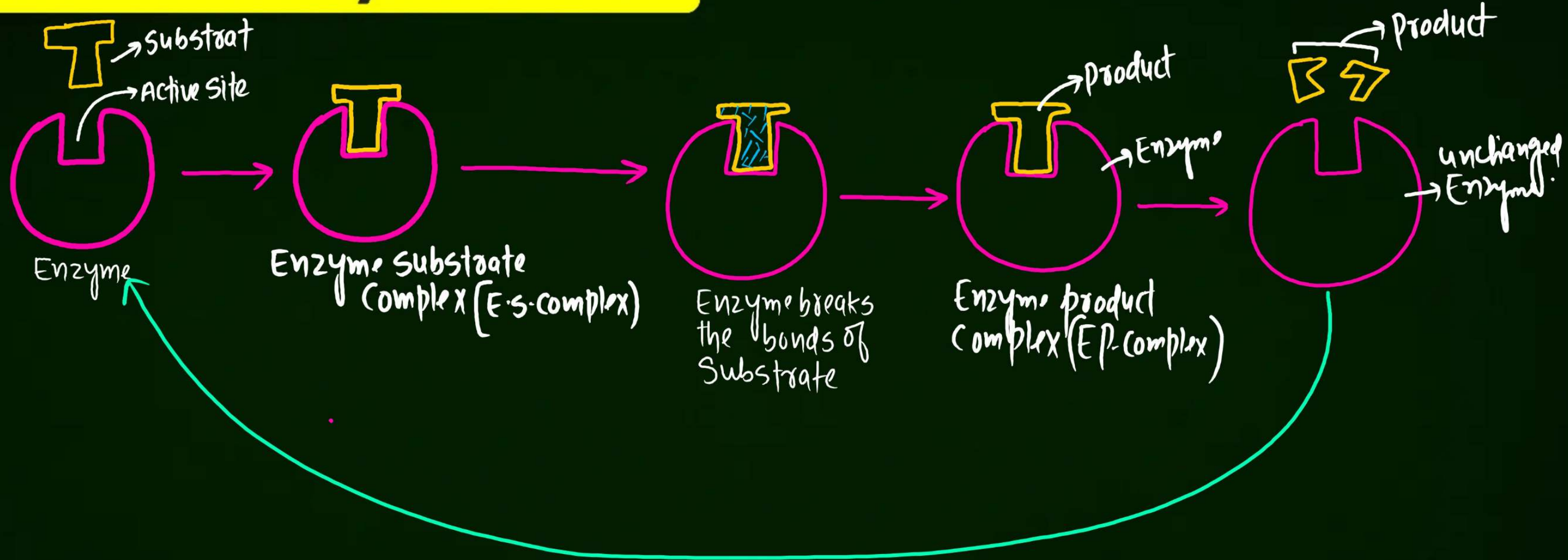


How Enzyme Fastens a Chemical Reaction?

- The substrate (S) binds to a specific part of the enzyme called the active site.
- This forms a temporary enzyme-substrate (ES) complex.
- While bound, the substrate changes into a transition state (unstable, high-energy state).
- After bond changes are done, the product (P) is released.
- The reaction may involve several short-lived, unstable states before the final product forms.
- These states relate to energy levels, the transition state is the highest.
- The external energy needed by a substrate to start a chemical reaction & convert into product is called as activation energy.
- Enzymes lower this activation energy, making the reaction faster and easier.

Nature of Enzyme action

→ lock & key theory.



Nature of Enzyme action

The catalytic cycle of an enzyme action can be described in the following steps:

- a) Substrate binds to the enzyme's active site
- b) Enzyme changes shape to fit the substrate more tightly and forms Enzyme substrate Complex.
- c) Enzyme breaks substrate's chemical bonds and forms enzyme-product complex.
- d) Products are released; enzyme is free to repeat the cycle with another Substrate.

Question [NEET-2024]

Kaun Banayga Nabkaar

Regarding catalytic cycle of an enzyme action, select the correct sequential steps

- ② A. Substrate enzyme complex formation.
- ⑤ B. Free enzyme ready to bind with another substrate.
- ④ C. Release of products.
- ③ D. Chemical bonds of the substrate broken.
- ① E. Substrate binds to active site.

Choose the correct answer from the options given below.

~~1~~ E, A, D, C, B.

~~3~~ B, A, C, D, E

~~2~~ A, E, B, D, C

~~4~~ E, D, C, B, A

Kaun Banayga Nabkaar : NEET-2016

8. marks.

A non- proteinaceous enzyme is:

A Lysozyme X

B Ribozyme ✓

C Ligase Y

D Deoxyribonuclease X



Thank
You